XIAMEN UNIVERSITY MALAYSIA

ONLINE MID-TERM TEST 2

Course Code: COM107

Course Name: Circuits and Devices

Question Paper Setter: Chua Ming Yam

Academic Session: 2022/04

Total No. of Pages: 2 Time Allocated: 1 hour

Additional Materials: All materials related to COM107.

INSTRUCTIONS TO CANDIDATES

1. This paper consists of **TWO** pages excluding the cover page.
2. There are **TWO** questions in this paper. Attempt **ALL**.
3. Show all your calculations in your answer booklet.
4. Please type in all the corresponding answers on the downloadable answer book provided.
5. Communication between candidates by any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates or suspected of doing so, you would be penalized accordingly.

(Student ID: _____ Full Name: _____)

Question 1 (50 marks, CLO2-PLO2:C4)

A series RLC circuit has a $20\,\Omega$ non-reactive resistor, a coil of inductance $0.1\,H$ with negligible resistance, and a $470\,\mu F$ capacitor, as shown in Figure Q1. The circuit is supplied with an a.c. source, V, with an instantaneous voltage of,

$$v(t) = 240\cos\left(314.16t + \frac{\pi}{3}\right)$$

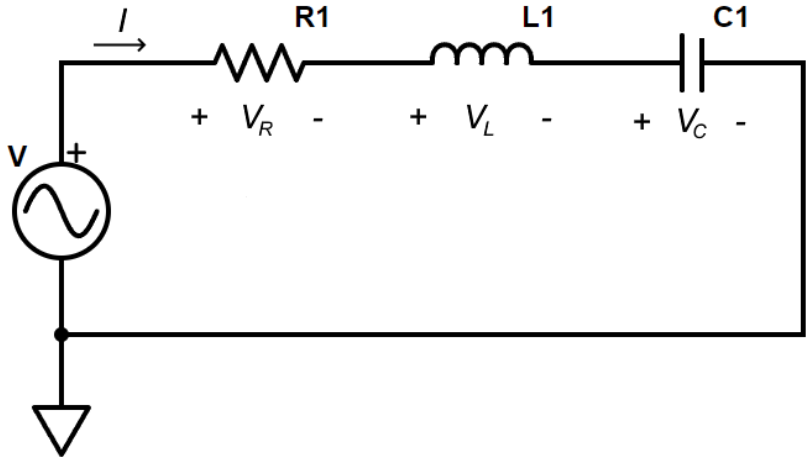


Figure 1

a) Derive the total impedance, Z_T, of the series RLC components. Express it in complex form. [4 marks]

b) Determine the current I. Express the current in complex form. [6 marks]

c) Determine the magnitude of the current, $|I|$, and the phase angle of the circuit. [6 marks]

d) Determine the voltage V_R, V_L, and V_C. Express them in complex form. [3 x 5 marks]

e) Sketch the phasor diagram for I, V, V_R, V_L, and V_C. Indicate the phase angle of all the components. [10 marks]

f) If the coil has a resistance of 3 Ω, determine the phase angle of the circuit. [9 marks]

** The phase angle of a circuit is the phase difference between the voltage and the current.*

Continues…

Question 2 (50 marks, CLO2-PLO2:C4)

A parallel RLC circuit has a $20\ \Omega$ non-reactive resistor, a coil of inductance of $0.1\ H$ with negligible resistance, and a $470\ \mu F$ capacitor, as shown in Figure Q2. The circuit is supplied with an a.c. source, V, with an instantaneous voltage of,

$$v(t) = 240\cos\left(314.16t + \frac{\pi}{3}\right)$$

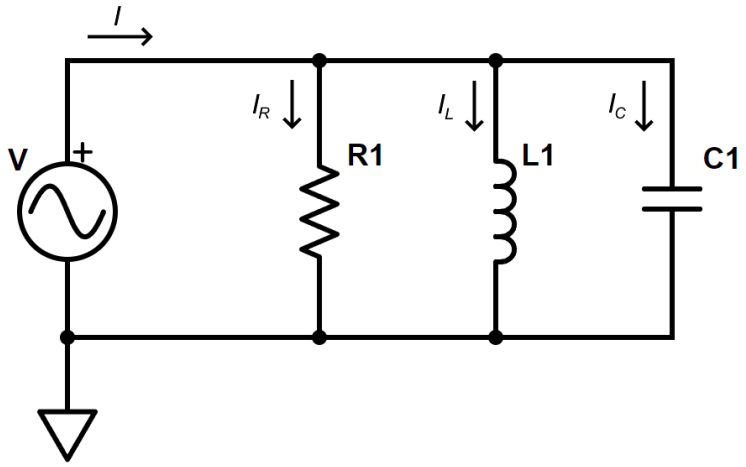


Figure Q2

a) Derive the total impedance, Z_T, of the parallel RLC components. Express it in complex form. [4 marks]

b) Determine the current I. Express the current in complex form. [6 marks]

c) Determine the magnitude of the current, $|I|$, and the phase angle of the circuit. [6 marks]

d) Determine the voltage I_R, I_L, and I_C. Express them in complex form. [3 x 5 marks]

e) Sketch the phasor diagram for V, I, I_R, I_L, and I_C. Indicate the phase angle of all the components. [10 marks]

f) Tune the circuit's phase angle to -45°. [9 marks]

** The phase angle of a circuit is the phase difference between the voltage and the current.*

- END OF PAPER -